

# IF440: Artificial Intelligence

## Topic 06: Markov Chains

Moeljono Widjaja, Ph.D.

Yaya Suryana, Ph.D.

**Universitas Multimedia Nusantara**  
Jl. Scientia Boulevard, Gading Serpong  
Tangerang, Banten-15811 Indonesia

Odd Semester 2025-2026

MULTIMEDIA  
NUSANTARA

## Course Sub Learning Outcomes (Sub-CLO):



SubCLO-0221: Students are able to apply knowledge-based reasoning, probabilistic reasoning and soft computing (C3).



# Review

- 1 Probabilistic Reasoning
- 2 Bayesian Network
- 3 Bayesian Network Inference



# Outline

- 1 Markov Chain
- 2 Transition Matrix
- 3 Example of Applied Markov Chains
- 4 Practices



UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# Pre-Class Assignments



- Dr. Trefor Bazett <https://www.youtube.com/watch?v=rHdX3ANxofs>
- Dr. Trefor Bazett <https://www.youtube.com/watch?v=1GKtfgwf3ig>

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# Markov Chain

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# WHAT IS A MARKOV CHAIN?

- A **Markov chain** is a mathematical system that experiences transitions from one state to another according to certain probabilistic rules.
- **Used to** compute the probabilities of events occurring by viewing them as states transitioning into other states, or transitioning into the same state as before.

# WHAT IS A MARKOV CHAIN?

- One special type of discrete-time is called a Markov Chain.
- **Definition:** A discrete-time stochastic process is a Markov Chain if, for  $t = 0, 1, 2, 3, \dots$  and all states

$$P(X_{t+1} = i_{t+1} \mid X_t = i_t, \dots, X_1 = i_1, X_0 = i_0) = P(X_{t+1} = i_{t+1} \mid X_t = i_t)$$

- Essentially this says the probability distribution of the state at time  $t+1$  depends on the state at time  $t(i_t)$  and does not depend on the states the chain passed through on the way to  $i_t$  at time  $t$ .



# WHAT IS A MARKOV CHAIN?

- We have a set of states,  $\mathbf{S} = \{s_1, s_2, \dots, s_r\}$ .
- The process starts in one of these states and moves successively from one state to another.
- Each move is called a **step**.

# WHAT IS A MARKOV CHAIN?

- Further assumptions that for all states  $i$  and  $j$  and all  $t$ ,  $P(X_{t+1} = j \mid X_t = i)$ , is independent of  $t$ .
- This assumptions allows us to write  $P(X_{t+1} = j \mid X_t = i) = p_{ij}$  where  $p_{ij}$  is the probability that the given system is in state  $i$  at time  $t$ , it will be in state  $j$  at time  $t+1$ .
- If the system moves from state  $i$  during one period to state  $j$  during the next period, we call that a transition from  $i$  to  $j$  has occurred.



## Transition Matrix

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# TRANSITION MATRIX

- $p_{ij}$ 's are often referred to as the **transition probabilities** for the Markov chain.
- We also must define  $q_i$  to be the probability that the chain is in state  $i$  at the time 0; in other words  $P(X_0 = i) = q_i$ .

# TRANSITION MATRIX

- In most applications, the transition probabilities are displayed as an  $s \times s$  **transition probability matrix**  $P$ . The transition probability matrix  $P$  may be written as

$$P = \begin{bmatrix} P_{11} & \cdots & P_{1s} \\ P_{21} & \cdots & P_{2s} \\ \vdots & \vdots & \vdots \\ P_{s1} & \cdots & P_{ss} \end{bmatrix}$$

- For each  $i$

$$\sum_{j=1}^s p_{ij} = 1$$

# n-STEP TRANSITION PROBABILITIES

- A question of interest when studying a Markov Chain: If a Markov chain is in a state  $i$  at time  $m$ , what is the probability that  $n$  periods later the Markov chain will be state  $j$ ?

$$P(X_{m+n} = j \mid X_m = i) = P(X_n = j \mid X_0 = i) = P_{ij}(n)$$

where is called the  $n$ -step probability of a transition from state  $i$  to  $j$ .

# n-STEP TRANSITION PROBABILITIES

- For  $n > 1$ ,  $P_{ij}(n)$  =  $ij^{\text{th}}$  element of  $P^n$
- $P_{ij}(2)$  is the  $(i,j)^{\text{th}}$  element of the matrix  $P^2 = P^1 P^1$
- $P_{ij}(n)$  is the  $(i,j)^{\text{th}}$  element of the matrix  $P^n = P^1 P^{n-1}$



# EXAMPLE

- Suppose the entire cola industry produces only two colas. Given that a person last purchased cola 1, there is a 90 % chance that their next purchase will be cola 1. Given that a person last purchased cola 2, there is an 80% chance that their next purchase will be cola 2.
  1. Find  $P(X_1=\text{cola 2} \mid X_0=\text{cola 1})$
  2. If a person is currently a cola 2 purchaser, what is the probability that they will purchase cola 1 two purchases from now?
  3. If a person is currently a cola 1 purchaser, what is the probability that they will purchase cola 1 three purchases from now?

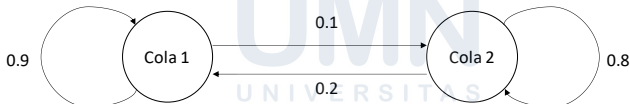


# EXAMPLE

## Transition matrix

$$P = \begin{matrix} & \begin{matrix} cola\ 1 & cola\ 2 \end{matrix} \\ \begin{matrix} cola\ 1 \\ cola\ 2 \end{matrix} & \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \end{matrix}$$

## Transition diagram



# EXAMPLE

$$P = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}$$

We can now answer questions 1 and 2

1.  $P(X_1 = \text{cola 2} \mid X_0 = \text{cola 1}) = 0.1$
2.  $P(X_2 = 1 \mid X_0 = 2) = P_{21}(2)$

$$P^2 = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix}$$

Hence,  $P_{21}(2) = 0.34$

$$2. \quad P_{11}(3) = P^3 = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix} \begin{bmatrix} 0.83 & 0.17 \\ 0.34 & 0.66 \end{bmatrix} = \begin{bmatrix} 0.781 & 0.219 \\ 0.438 & 0.562 \end{bmatrix}$$

Hence,  $P_{11}(3) = 0.781$



## Example of Applied Markov Chains

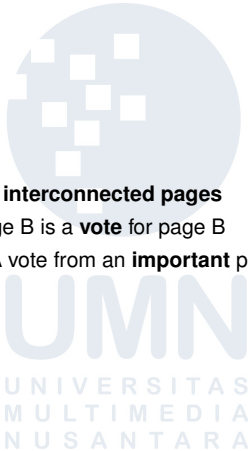
UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# Applied Markov Chains: PageRank

- Developed by **Larry Page and Sergey Brin** (1996)
- Used by **Google Search** to rank web pages
- Based on **Markov Chains and probability theory**

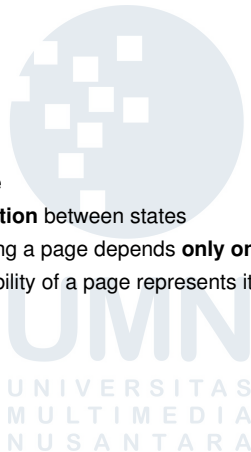


# Key Idea Behind PageRank

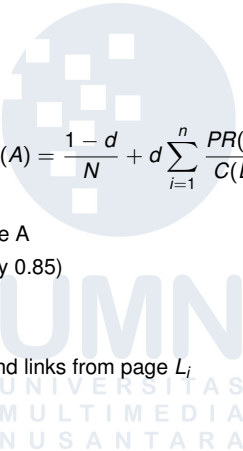
- 
- The web is a **network of interconnected pages**
  - A link from page A to page B is a **vote** for page B
  - Not all votes are equal: A vote from an **important** page is more valuable

# Markov Chain Model

- Each webpage is a **state**
- Clicking a link is a **transition** between states
- The probability of reaching a page depends **only on the current page**
- The **steady-state** probability of a page represents its importance

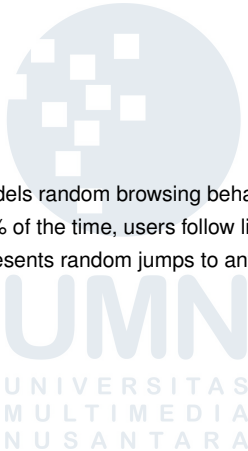


# PageRank Formula


$$PR(A) = \frac{1-d}{N} + d \sum_{i=1}^n \frac{PR(L_i)}{C(L_i)} \quad (1)$$

- $PR(A)$ : PageRank of page A
- $d$ : Damping factor (usually 0.85)
- $N$ : Total number of pages
- $L_i$ : Pages linking to A
- $C(L_i)$ : Number of outbound links from page  $L_i$

# Damping Factor Explanation

- 
- **Damping Factor (d)** models random browsing behavior
  - **Typical value: 0.85** (85% of the time, users follow links)
  - The remaining 15% represents random jumps to any page



# Example Calculation

- Consider 3 pages: A, B, C
- Links:  $A \rightarrow B$ ,  $B \rightarrow C$ ,  $C \rightarrow A$
- Initial ranks:  $PR(A) = PR(B) = PR(C) = \frac{1}{3}$
- Iterative updates using the PageRank formula



# Convergence and Iteration

- PageRank values are **computed iteratively**
- Algorithm stops when ranks **converge** (small changes between iterations)
- Typically, 20-50 iterations are needed for convergence

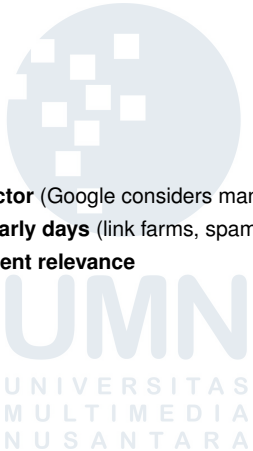


# Importance in SEO

- Higher PageRank = **Higher visibility in search results**
- Strategies to improve PageRank:
  - **Get backlinks from authoritative sites**
  - **Avoid link spamming** (Google penalizes manipulation)
  - **Improve internal linking** structure

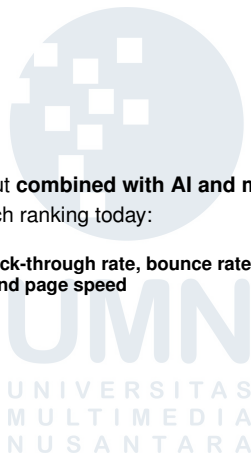


# Limitations of PageRank

- 
- **Not the only ranking factor** (Google considers many factors now)
  - **Easily manipulated in early days** (link farms, spam)
  - **Does not consider content relevance**

# Modern Google Ranking

- PageRank is still used but **combined with AI and machine learning**
- Factors influencing search ranking today:
  - **Content quality**
  - **User engagement (click-through rate, bounce rate)**
  - **Mobile-friendliness and page speed**





## Practices

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# PRACTICE 1

Consider a Markov chain with three possible states 1, 2, and 3 and the following transition probabilities matrix:

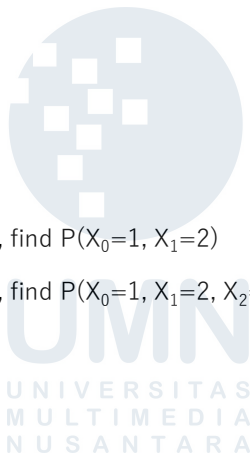
$$P = \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 0 & 1/2 \end{bmatrix}$$

Draw the transition diagram of the given matrix.

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# PRACTICE 1

1. Find  $P(X_4=3 \mid X_3=2)$
2. Find  $P(X_3=1 \mid X_2=1)$
3. If we know  $P(X_0=1) = 1/3$ , find  $P(X_0=1, X_1=2)$
4. If we know  $P(X_0=1) = 1/3$ , find  $P(X_0=1, X_1=2, X_2=3)$





## PRACTICE 2

- The Land of Oz is blessed by many things, but not by good weather. They never have two nice days in a row. If they have a nice day, they are just as likely to have snow as rain the next day. If they have snow or rain, they have an even chance of having the same the next day. If there is change from snow or rain, only half of the time is this a change to a nice day. With this information we form a Markov chain as follows.
- We take as states the kinds of weather R, N, and S. From the above information we determine the transition probabilities. These are most conveniently represented in a square array as

UMN  
UNIVERSITAS  
MULTIMEDIA  
NUSANTARA

# PRACTICE 2

1. Make the transition matrix.
2. Draw the transition diagram of the matrix.
3. If today is raining, what is the probability that tomorrow will be snowing?
4. If today is snowing, what is the probability that 2 days from now will be a nice weather?

# References

- Weber, R. (2011), Markov Chains, University of Cambridge.  
<http://www.statslab.cam.ac.uk/~rrwl/markov/M.pdf>



# Next Week

- What is an HMM?
- HMM Formalism
- Example
- Viterbi Algorithm



# Vision and Mission

## Vision

Becoming a leading Informatics Undergraduate Study Program that produces graduates with international insight who are competent in the field of computer science, have an entrepreneurial spirit, and have noble character.



## Mission

- 1 Organising learning with the best technology and curriculum and supported by professional teaching staff.
- 2 Carry out research activities in the field of informatics to advance information science and technology.
- 3 Carry out community service activities based on information technology and science in the context of practising informatics science and technology society.

